

3 (a) (i) A field of force is a region of space where a body experiences a force due to the presence of another of similar type.

(ii) The denser and closer the field lines are to one another, the greater the strength of the force field.

(b) (i) Mass

(ii) Positive charge

(iii) Nothing.

(c) The electric field and magnetic field exerts a force qE and Bqv respectively in the opposite direction. Only in the case of charged particles with a particular speed v will these two forces cancel out so that particles pass through undeflected.

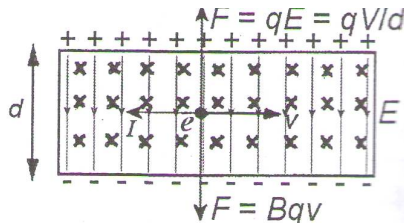
The condition is that

Electric force = magnetic force

$$qE = Bqv$$

$$v = \frac{E}{B}$$

This is illustrated below:



Note: Seldom was direction of electric field made clear. Both B and E field directions must be shown clearly. State clearly forces are not only equal in magnitude but also opposite in direction.

4 (a) $R_1 + R_2 + R_3 = 12 \Omega$